



OUTDOOR ROOMBA

User Testing

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Mitchell Owen: *Electrical Engineer, Tester*

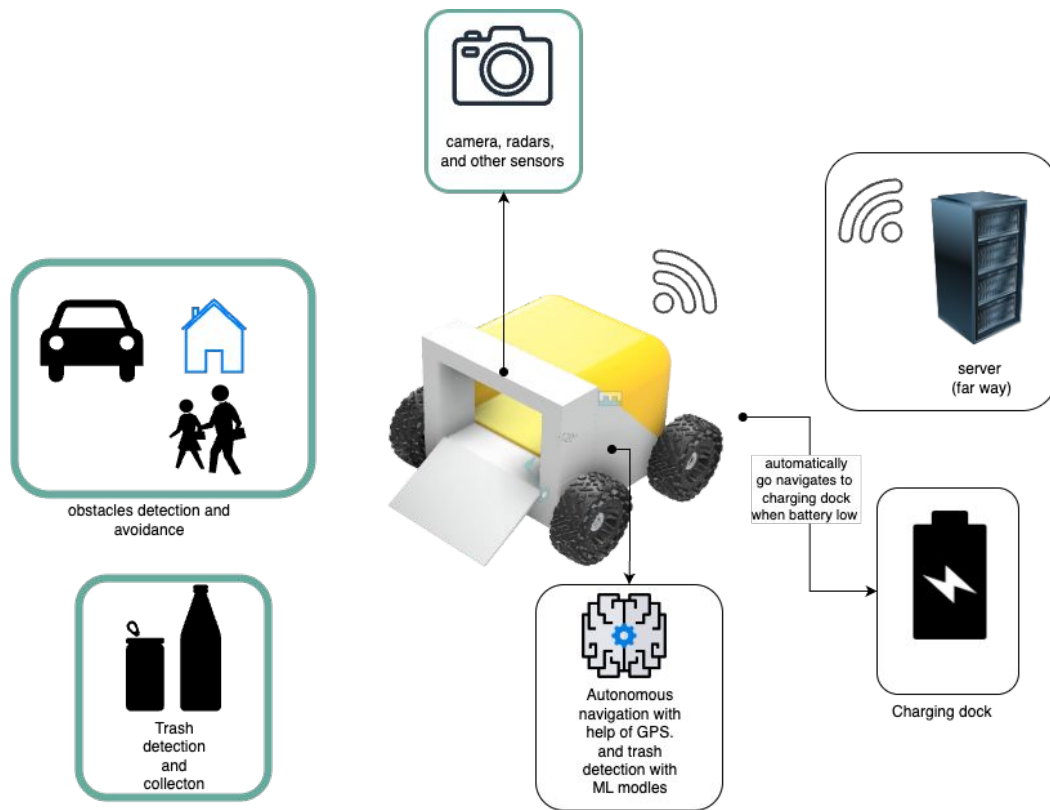
Om Shukla: *Backend Engineering*

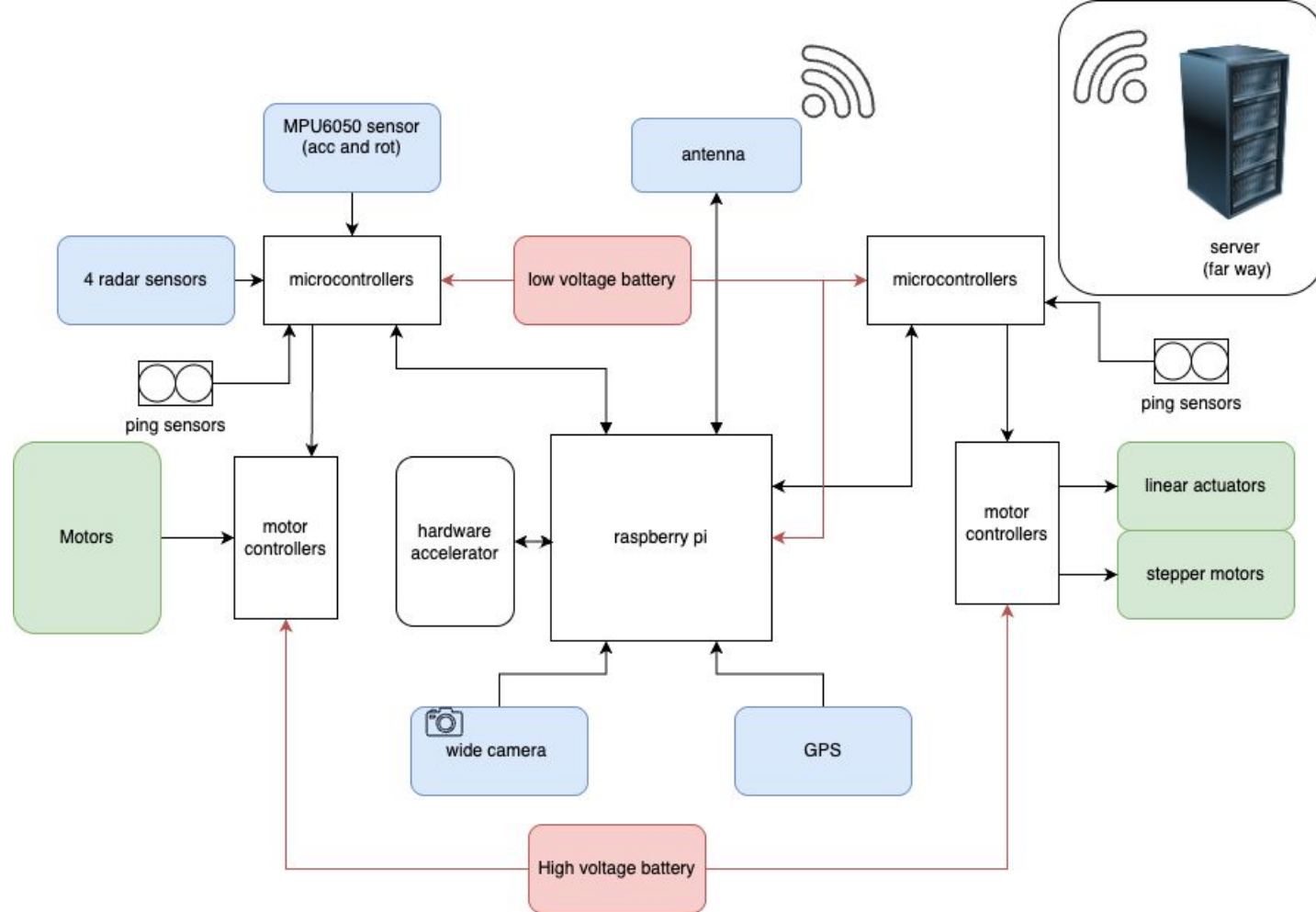
Daniel Vergara Pinilla: *Machine Learning Engineer*

Overview

Outdoor Roomba

1. Autonomous
2. Object and Trash recognition
3. Refuse Collector
4. Unified Endpoint Management





Survey

- Pre-Testing: Getting to know the users.
- Scenario: How will they use our product.
- Post-Testing: How Satisfied are the Users with our product.

Outdoor Roomba Usability Test

* Indicates required question

Name of Person / Company

Do you currently use any other autonomous cleaning devices? (If yes please state * name of product. If no, please state N/A)

Your answer

If you answered yes to the last question, please rate from 1-10 on how satisfied you are with your current model. *

1 2 3 4 5 6 7 8 9 10

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

Do you ever find that you don't have enough time to keep your place clean? * Explain answer.

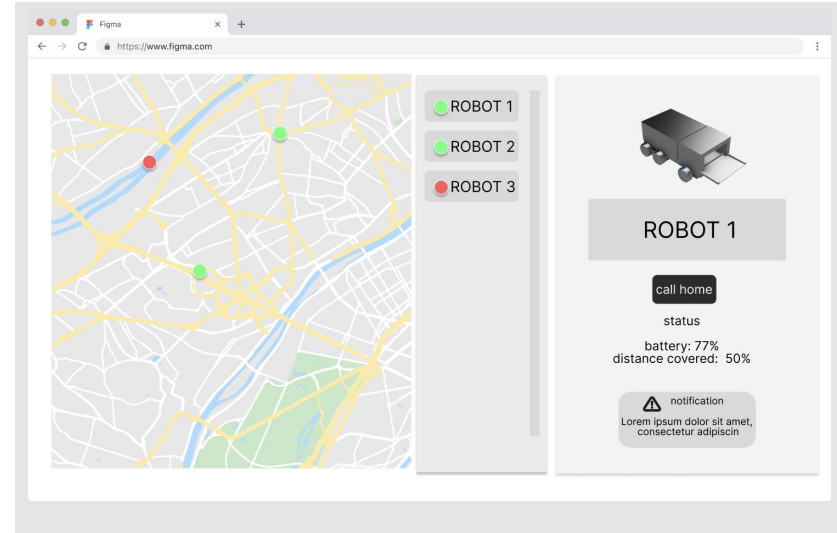
Your answer

How often do you or your company throw events? *

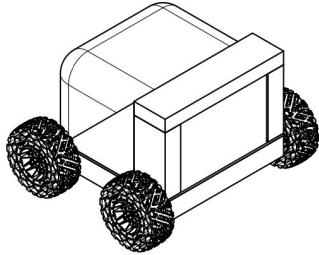
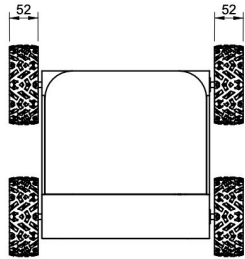
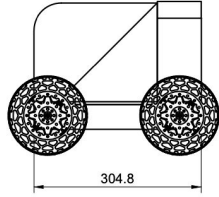
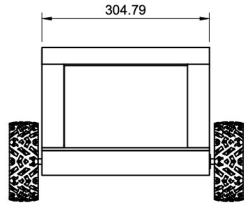
- ☐ Once a year.
- ☐ Multiple Months out of the year
- ☐ Multiple Weeks
-

Prototype Evaluation

- Have users test and evaluate an early prototype of the robot
- Users would complete tasks that we have set up.
- Questions and tasks would change based on the prototype
- Used to evaluate the usability of the robot and application



Concept of User Interface (final product will not look like this at all)



Prototype without linear actuators

User Testing / Results

Based on the results from the testing we will modify and improve our robot to better fit the needs of the user.

Possible Modifications:

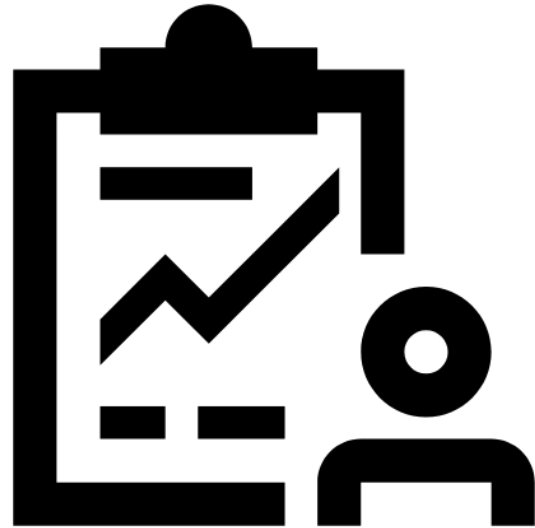
- Battery Life
- Storage Capacity
- Ease of Use
- Sensor adjustments
- Machine learning models

User Reports

Have users that own the product fill out a survey or report

Report will contain:

- How long the roomba is in use daily
- What types of trash get picked up
- How frequently does the roomba fill up
- How many stops the roomba makes



Questions

